appendix B
Ingest pipelines

Data that makes its way into Elasticsearch is not always clean. Usually, data requires transformation, enrichment, or formatting. There are options for cleaning data before bringing it into Elasticsearch for ingestion, such as writing custom transformers or using ETL (extract, transform, load) tools. Elasticsearch allows these capabilities via ingest pipelines that provide first-class support for manipulating data—we can split, remove, modify, and enhance data before it is ingested.

B.1 Overview

Data to be indexed into Elasticsearch may need to undergo transformation and manipulation. Consider an example of loading millions of legal documents represented as PDF files into Elasticsearch for searching. Although bulk loading them is one approach, it is inadequate, cumbersome, and error-prone.

If you are thinking we can use ETL tools for such data-manipulation tasks, you are absolutely right. A plethora of such tools are available, including Logstash. Logstash manipulates our data before it is indexed into Elasticsearch or persisted to a database or other destinations. However, it is not lightweight, and it requires an elaborate setup, preferably on a different machine.

Just as we write search queries using Query DSL, we can develop ingest pipelines with processors using the same syntax and apply them to incoming data to ETL the data. The workflow is straightforward:

1. Create one or more pipelines with the expected logic based on the business requirements regarding transformations, enhancements, or enrichments to be carried out on the data.
2. Invoke the pipelines on the incoming data. The data goes through the series of processors in a pipeline and is manipulated at every stage.
3. Index the processed data.